Chronic Cutaneous Lupus Erythematosus

Most patients with untreated classic discoid lupus erythematosus (DLE) lesions experience indolent progression to extensive cutaneous dystrophy and scarring alopecia, which can be psychosocially devastating and occupationally disabling. However, with appropriate treatment, skin disease can be largely controlled. Spontaneous remission occurs occasionally, but disease activity may recrudesce at sites of older, inactive lesions. Rebound flares after discontinuation of therapy are common, and a slow taper of medications during periods of disease inactivity is recommended. Squamous cell carcinoma may rarely develop in long-standing, chronic DLE lesions.

Death from systemic lupus erythematosus (SLE) is uncommon in patients who initially present with localized DLE. As noted in epidemiologic studies, these patients have only a 5% risk of developing clinically significant SLE. Generalized DLE and persistent, low-grade laboratory abnormalities appear to increase the risk of progression to systemic disease. Additionally, unrecognized squamous cell carcinoma arising within a long-standing DLE lesion may contribute to morbidity and mortality.

Outcomes Measures

The development of the Cutaneous Lupus Erythematosus Area and Severity Index (CLASI) has enabled objective assessment of CLE disease activity and damage over time. The CLASI includes separate scores for disease activity and damage, which is critical because scarred or "burned-out" areas are not expected to improve with therapies targeting active inflammation. This tool is particularly timely given the U.S. Food and Drug Administration's call for new drug development for lupus, including forms limited to a single organ such as skin-predominant DLE.

A Brazilian research group has utilized the Dermatology Life Quality Index (DLQI) and the SF-36 health survey to assess quality of life in patients with CLE. Both instruments demonstrated that active skin lesions significantly impair quality of life, with patients who also have alopecia experiencing particularly severe impacts.

TREATMENT

Initial management of any form of chronic cutaneous lupus erythematosus (CLE) should include evaluation to exclude underlying SLE at diagnosis. All patients should receive counseling on photoprotection, including avoidance of sunlight and artificial ultraviolet radiation (UVR), and should be advised to avoid potentially photosensitizing medications such as hydrochlorothiazide, tetracycline, griseofulvin, and piroxicam.

Specific therapy should prioritize optimization of local treatments, with systemic agents reserved for cases of persistent significant local disease activity or coexisting systemic involvement.

Acute cutaneous lupus erythematosus (ACLE) lesions typically respond to systemic immunosuppressive therapies used to manage associated SLE (e.g., systemic glucocorticoids, azathioprine, cyclophosphamide). Evidence increasingly supports the use of aminoquinoline antimalarials like hydroxychloroquine, which can exert a steroid-sparing effect in SLE and may benefit ACLE. Local therapies (discussed below) may also be helpful in ACLE.

In contrast, subacute cutaneous lupus erythematosus (SCLE) and chronic cutaneous lupus erythematosus (CCLE) often occur in patients without significant systemic disease. Therefore, nonimmunosuppressive approaches are preferred for these forms (see Table 156-7). In general, SCLE and CCLE lesions respond similarly to first-line local and systemic therapies.

Local Therapy

SUN PROTECTION

Patients should be advised to: - Avoid direct sun exposure, especially during peak UV hours - Wear tightly woven clothing and broad-brimmed hats - Apply broad-spectrum, water-resistant sunscreen regularly (SPF $\geq$ 30) containing effective UVA blockers such as: - Photostabilized avobenzone (Parsol 1789) - Micronized titanium dioxide - Micronized zinc oxide - Mexoryl SX

Additional protective measures include: - Installing UV-blocking films on home and automobile windows - Placing acrylic diffusion shields over fluorescent lights

Corrective camouflage cosmetics (e.g., Dermablend®, Covermark®) provide dual benefits: they act as effective physical sunscreens and offer cosmetic coverage.

For a comprehensive review of photoprotection and local management strategies in autoimmune connective tissue skin diseases, refer to Chapter 223 by Ting and Sontheimer.